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RESEARCH ARTICLE

Real-Time Detection Method for Cracked Eggs Based on Improved YOLOv8 and Dual Verification With Triple Classification Granularity

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ABSTRACT Addressing the urgent need for real-time detection of cracked eggs on production lines, this study proposes a high-precision real-time detection algorithm based on an improved YOLOv8 framework. This study innovatively introduced a dual-detection method, incorporating a ''crack region'' classification to assist in distinguishing between intact eggs and cracked eggs. This method identifies cracked eggs by recognizing either local or overall crack characteristics, effectively reducing both false negatives and false positives. At the architectural level, the model innovatively integrates Weighted Bidirectional Feature Pyramid (BiFPN) to optimize multi-scale feature fusion, Deformable Convolutional Network v2 (DCNv2) to enhance perception of irregular geometric deformations in cracks, Weighted Intersection over Union v2 (WIoUv2) loss function to improve detection accuracy for occluded samples, and adds a small object detection layer to strengthen recognition of minute defects. Experiments demonstrate that the improved model achieves a mean average precision (mAP) of 92.9% on the test set, representing a 5.2% improvement over the original model and outperforming current mainstream detection models. In real-world production deployment testing, the system achieved a real-time detection speed of 45 FPS, with a false negative rate of only 0.46% and a false positive rate of 2.18% for intact eggs, while keeping the false positive rate for broken eggs below 2%. This performance far surpasses that of the original model and fully meets the requirements for industrial-grade real-time online quality inspection. This study provides an efficient solution for egg quality inspection with significant engineering application value.

INDEX TERMS Cracked eggs, deep learning, real-time detection, improved YOLOv8, image processing.

I. INTRODUCTION

On the automated egg production line, the rapid and accurate identification of egg shell damage is a key technical link to ensure product quality and reduce enterprise losses [1], [2], [3]. About 3% of eggs have their shells damaged due to squeezing and impact during egg production and transportation [4]. Currently, the industry in China primarily relies on manual inspection, which faces issues of low efficiency and suboptimal accuracy [5]. Existing techniques include acoustic response and computer vision. Acoustic methods are prone to environmental interference and high costs [6], while vision-based approaches, with their non-contact and low-cost advantages, have become a research focus [7], [8].

Some researchers have employed traditional image processing techniques for cracked egg detection. For instance, Liang et al. achieved a 98.18% recognition accuracy using threshold segmentation based on histograms and the Otsu method [9]. Yao et al. improved the detection accuracy of scattered yolk eggs to 97.33% through image enhancement and morphological feature analysis [10]. However, traditional image processing methods require manual feature extraction,

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resulting in slow detection speeds and limited generalization capabilities. In recent years, deep learning has provided new approaches for object detection [11], [12]. Datta et al. achieved a mean average precision (mAP) of 75% for detecting severely cracked eggs using Faster R-CNN, whilst precisely localising the damaged areas [13]. Turkoglu's CNN-BiLSTM hybrid model further improved accuracy to 96.3% [14]. Botta et al. employed convolutional neural networks (CNNs) to detect cracked eggs, achieving an accuracy of 95.38% [15]. In domestic research, Li et al. achieved 99% accuracy using the ShuffleNet model [16]. Qi et al. combined CNN feature extraction with SVM classification, achieving an accuracy of 97.97% [17]. Regarding model optimization, Nasiri et al. reached an average accuracy of 94.84% via an improved VGG16 model with five-fold cross-validation [7]. Tang et al. reported a 96.3% accuracy for cracked preserved egg detection using an optimized MobileNet V3_large model [18]. Furthermore, Xiao et al. achieved an overall accuracy of 95.88% with an improved GoogLeNet-Mini network [19]. Chen et al. achieved a comprehensive accuracy of 96.7% using a transfer learning model based on EfficientNet [20]. Yao et al. further improved accuracy by 1.74%-4.31% by integrating an improved CNN with a hierarchical SVM [21]. However, most of these studies focus on optimizing for static images, suffer from high network complexity, or rely on serial processing mechanisms, resulting in inference speeds that struggle to meet the real-time demands of production lines.

Currently, research on real-time detection of cracked eggs within actual production line environments remains limited, despite such scenarios demanding stringent requirements for both detection accuracy (>99%) and speed (≥30 FPS). The YOLO (You Only Look Once) series of algorithms, leveraging their single-stage detection architecture and highly efficient parallel computing capabilities, achieve high efficiency in real-time detection. This makes them suitable for dynamic application scenarios with high-speed requirements [22]. In recent years, significant progress has been made in research improving the YOLO series of algorithms. For example, Yin et al. introduced Ghost modules and an ECA attention mechanism into YOLOv5, achieving a 93.8% accuracy for cracked duck egg detection [23]. Yao and Li et al. created a lightweight YOLOv5 model by integrating MobileNetv3, reducing computational cost by 35% while maintaining over 95% detection precision [24]. Huang et al. integrated CA attention mechanism, BiFPN, and GSConv modules, enabling their YOLOv5 model to achieve 96.4% accuracy for cracked egg detection in dynamic scenes [25]. Yang et al. optimized anchor boxes via K-means clustering and replaced the Hardswish activation function, achieving a mean Average Precision of 98.3% for cracked pigeon egg detection with YOLOv5 [26].

Although the aforementioned studies have yielded certain results in avian egg defect detection, several critical issues remain unresolved. First, most methods lack a rotational mechanism to present the entire egg-shell, leading to false

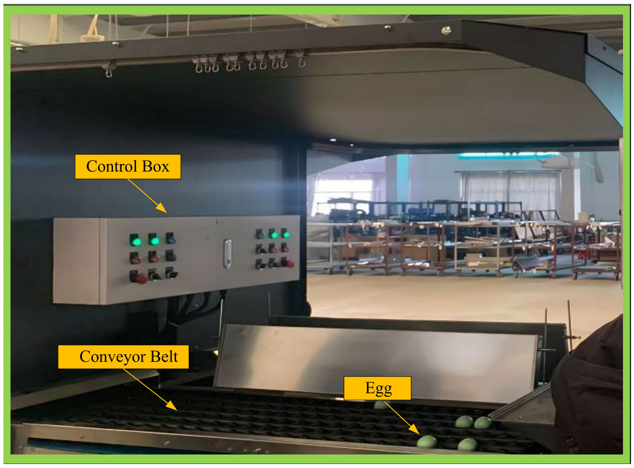


FIGURE 1. Egg conveying device.

negatives when cracks are located on the bottom. Second, existing algorithms are primarily designed for static images, making them inadequate for the high-speed motion and complex lighting conditions on production lines. Finally, in large-scale detection scenarios, the model's detection speed and accuracy still require optimisation. To address these issues, this paper takes cracked eggs collected by Guangzhou Suiping Enterprise Co., Ltd. as its research subject. Building upon the YOLOv8 detection model, it proposes an improved YOLOv8 cracked egg detection method tailored for dynamic production line environments. The main contributions of this paper are as follows:

(a) This study collects dynamic images from actual production lines, establishes a finely labeled dataset with three categories, including intact eggs, broken eggs, and broken areas, and proposes a dual discrimination mechanism that requires simultaneous detection of broken areas in broken eggs, enhancing the reliability of crack recognition and the stability of industrial applications.

(b) An improved YOLOv8 model structure is proposed for small crack recognition under high-speed motion and complex lighting conditions on production lines. By introducing BiFPN, DCNv2, WIoUv2, and a small target detection layer, the model's crack area perception capability, multi-scale feature fusion, and handling of difficult cases are significantly enhanced.

(c) The model is successfully deployed on experimental platforms and real production lines, significantly improving detection performance. Test results show that the improved model has high accuracy, stability, and reliability, and its detection speed meets the requirements of industrial production line applications.

II. MATERIALS AND METHODS

A. COMPUTER VISION DETECTION SYSTEM

In this study, the device shown in Figure 1 was used to transport eggs, consisting of a conveyor belt and a control box. The control box regulates belt speed and the lights. The belt has six channels, each transporting 60 eggs per minute, totaling 360 eggs/minute. Rollers on the belt, driven by a motor, making the eggs perform the compound movement of translation and rotation on the conveyor belt.

On the basis of the conveying device, a machine vision system, as shown in Figure 2, is designed. comprising a computer, camera, light source, and egg conveyor. The camera plane parallels the egg motion plane to prevent distortion. As the egg rotates past the camera's field of view, the camera captures the entire surface of the egg. The algorithm is deployed on the detection system's computer for testing.

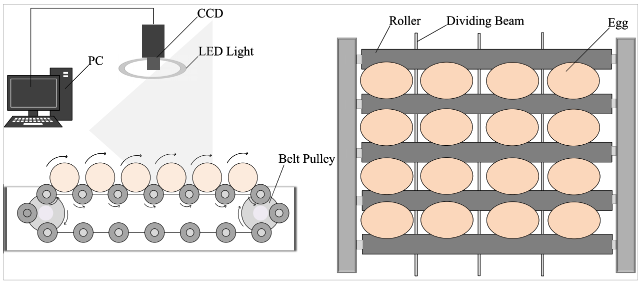


FIGURE 2. Proposed machine vision system.

This study integrates an improved YOLOv8 detection model into the system and employs a post-processing decision-making method based on dual verification. This decision-making method is based on three classifications of egg features: "intact eggs", "cracked eggs", and "crack regions". Under this post-processing method, the system classifies an egg as damaged whenever the model detects either a "cracked egg" or a "crack region".

B. DATA MATERIALS

1) IMAGES ACQUISITION

In order to make the dataset more suitable for actual working conditions, photos of unwashed eggs on the egg processing production line of Guangzhou Suibing Company were captured. This dataset contains 628 images, each of which includes one or more intact and cracked eggs to ensure the diversity of the sample. When photographing eggs on the production line, the researchers captured image data of eggs at various positions, angles, lighting conditions, and backgrounds along the actual production line. During this process, the researchers deliberately selected cracked eggs for photography to increase the proportion of cracked eggs in the images, thereby ensuring a balanced number of positive and negative samples.

2) DATASET PRODUCTION AND DIVISION

This study employs a dual-validation post-processing method based on the proposed triple classification granularity for dataset production and division. The study utilized the open-source annotation tool LabelImg to annotate the image dataset. The label categories comprised "1", "2", and "3", corresponding to "intact eggs", "cracked eggs", and "crack regions", respectively. This classification of egg features provides the model with richer information about damage characteristics, helping to enhance the model's perception and generalization capabilities. At the same time, compared to traditional binary classification (intact eggs vs. cracked eggs), the model can not only make judgments based on overall damage features but also incorporate features from localized cracked areas, thereby reducing the risk of missed detections that arise from relying solely on overall features for identification.

The annotated dataset was divided into a training set (377 images), a validation set (126 images), and a test set (125 images) in a 6:2:2 ratio. To enhance the model's generalization ability and robustness, data augmentation techniques were applied to the training set (377 images), including random cropping and random horizontal/vertical flipping with a 50% probability, among others. After data augmentation, the dataset was filtered to remove invalid data. The final training set contained 2,972 images. Combined with the validation set (126 images) and the test set (125 images), the dataset totaled 3,222 images, including 7,611 intact eggs, 2,711 cracked eggs, and 2,935 cracked areas. The model was trained using the 3,222 images to learn the various damage features of eggs in the images. Subsequently, the generated.xml annotation file was converted into a.txt file compliant with the YOLOv8 network requirements using a script.

C. DETECTION MODEL

YOLOv8 builds upon previous YOLO versions by introducing the latest backbone and neck architectures, enhancing feature extraction and object detection performance. It adopts an anchor-free split Ultralytics head, improving accuracy and efficiency in the detection process. Furthermore, YOLOv8 focuses on maintaining the optimal balance between accuracy and speed, making it more suitable for real-time object detection tasks, particularly in industrial assembly line scenarios. Therefore, this paper improves the model based on YOLOv8 to achieve real-time detection of cracked eggs on production lines.

1) YOLOv8 MODEL IMPROVEMENT METHODS

To enable the model to comprehensively learn and integrate both local details and global information within images, thereby enhancing detection accuracy to reduce false negatives and false positives while meeting real-time online detection requirements, this study implemented a series of improvements to the original YOLOv8 model. These enhancements include redesigning the loss function, adding a small object detection head and a deformable convolution network (DCNv2), as well as embedding a weighted bidirectional feature pyramid network (BiFPN). The improved model architecture is illustrated in Figure 3.

Improvements should be made in the following four aspects:

(a) WIoU Loss Function

WIoU offers three versions (v1, v2, v3) to accommodate different datasets and detection task requirements [27]. WIoU

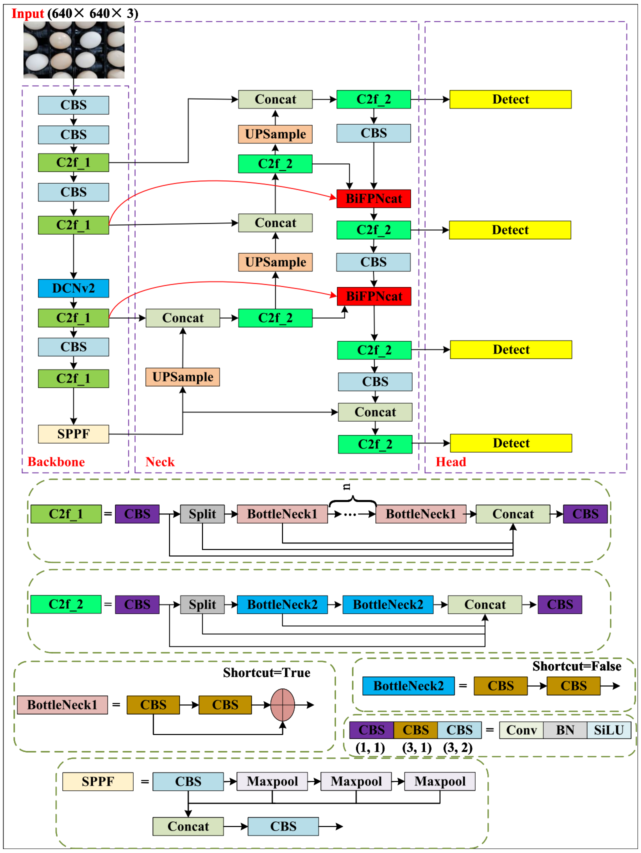


FIGURE 3. Improved network structure diagram of YOLOv8.

v2 dynamically adjusts the focus coefficient to adaptively balance gradient contributions from samples of varying quality, thereby accelerating model convergence and reducing computational overhead. Furthermore, during optimization, WIoU v2 comprehensively considers weight distribution between positive and negative samples, effectively mitigating class imbalance and enhancing the model's ability to distinguish challenging samples. Its core approach involves normalizing the IoU loss combined with an exponential adjustment factor, leading to a more stable and efficient training process. The calculation formula is given by:

$$L_{\mathrm{WIoUv2}} = L_{\mathrm{IoU}}^{\gamma^*} L_{\mathrm{WIoUv1}}, \gamma > 0. \tag{1}$$

where LWIoU v1 is the loss function of WIoU v1; LWIoU v2 is the loss function of WIoU v2; γ represents the modulation coefficient; $L_{\mathrm{IOU}}^{\gamma^*}$ is a dynamic scaling factor.

(b) Deformable Convolutions (DCNv2)

The fixed geometric structure of standard convolutional kernels limits their ability to effectively represent the features of objects with significant deformation or varying scales, which is a critical challenge in detecting irregular eggshell cracks. If it is assumed that the input feature image X and the standard square convolution receptive field are H, then each pixel q in the output feature image Y can be formalized as:

$$Y(q) = \sum_{k=1}^{K} w_k X (q + q_k). \tag{2}$$

where q_k represents the position in H, and w_k represents the weight of the sampled values at different positions. To overcome the limitation of fixed geometry, Deformable Convolutional Networks (DCNv1) introduced learnable spatial offsets $\{\Delta q_k \mid \mathrm{I}k = 1, \cdots, K\}$, enabling the sampling grid to adaptively deform towards the target's geometric structure [28]. Building upon this, the enhanced DCNv2 incorporates a modulation mechanism [29]. This mechanism not only learns spatial offsets but also assigns a dynamic modulation scalar Δm_k to each sampling point, which scales the feature amplitude. Thus, the formulation evolves to:

$$Y(q) = \sum_{k=1}^{K} w_k X (q + q_k + \Delta q_k) \Delta m_k. \tag{3}$$

where Δq_k and Δm_k denote the position offset and weight modulation parameter for the k-th sampling point, respectively. Both are obtained through a separate convolutional layer with the same spatial resolution and dilation rate as the current convolutional layer.

This dual mechanism allows DCNv2 to not only adjust the receptive field to align with irregular contours, such as eggshell cracks, but also to focus on more critical regions by amplifying the contributions of key sampling points while suppressing less informative ones. Therefore, the study replaces the standard convolutional layers in the backbone network of YOLOv8 with DCNv2 to enhance its capability in perceiving and representing fine and variable crack features. A comparative illustration of the sampling strategies for standard convolution, DCNv1, and DCNv2 is provided in Figure 4.

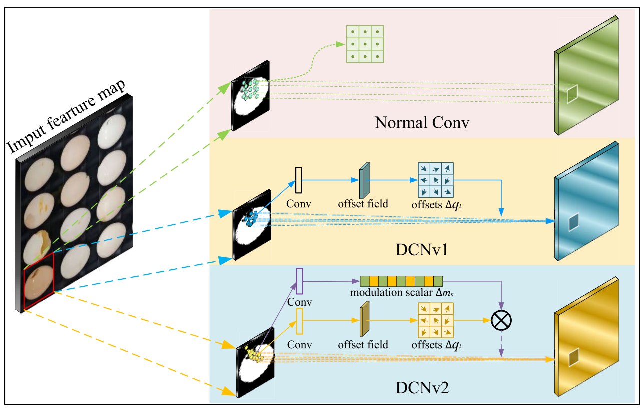


FIGURE 4. Comparison of DNv1, DCNv2 and Normal Conv sampling.

(c) BiFPN

BiFPN is an efficient multi-scale feature fusion network that builds upon the traditional Feature Pyramid Network (FPN). The standard FPN integrates high-level semantic features with low-level spatial features through a top-down pathway and lateral connections. PANet enhances this architecture by adding an extra bottom-up path, achieving bidirectional feature fusion. BiFPN further refines this process by introducing weighted feature fusion and employing a repeated block structure, which significantly improves fusion efficacy [30]. The damaged areas in cracked eggs can be relatively small with subtle features. By integrating both shallow and deep features, BiFPN generates more expressive feature maps, enabling the model to more accurately identify the characteristics of cracked eggs, thereby reducing false positive and false negative. Furthermore, the modular design of BiFPN allows the model to maintain high detection performance under complex scenarios such as varying lighting conditions and background clutter, which is crucial for the practical application of cracked egg detection. Therefore, in this study, the PANet structure in the YOLOv8 model is replaced with BiFPN. The architectures of FPN, PANet, and BiFPN are illustrated in Figure 5.

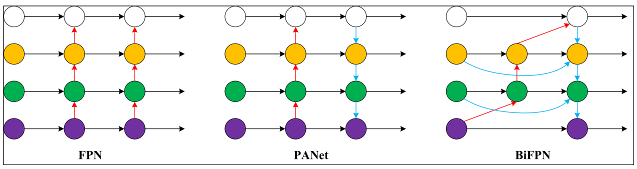


FIGURE 5. The structure of FPN, PANet, and BiFPN.

(d) Small Object Detection Layer

YOLOv8's suboptimal performance in detecting small objects is primarily due to the size constraints of small targets. By default, YOLOv8 processes input images at 640 × 640 resolution. and the backbone network performs five downsampling operations. The feature maps obtained by the detection heads are sized 80 × 80, 40 × 40, and 20 × 20, respectively. The 80 × 80 feature map is responsible for detecting small objects. When mapped onto the 640 × 640 input image, each cell in this feature map has a receptive field of 8×8 pixels. If an object in the input image has a length or height of less than 8 pixels, the network struggles to learn the target's feature information. To overcome this limitation and enhance detection accuracy, an additional detection layer specifically designed for small object detection can be employed. This facilitates multi-scale feature learning, enabling the model to leverage finer spatial details from earlier network stages and thereby improve its ability to recognize minute targets [31].

D. EXPERIMENTAL ENVIRONMENT AND PARAMETER CONFIGURATION

In this study, the experimental hardware configuration and software environment are as follows: the operating system is Windows 10, the programming language is Python 3.8, and the deep learning framework is PyTorch 1.13.1. The processor was an Intel(R) Core (TM) i5-10400F with a base frequency of 2.9GHz. The graphics card was an NVIDIA GeForce RTX 3060 with 12GB of VRAM. System memory was 32GB (3200MHz frequency), and the CUDA version was 11.6. During model training, input images were set to a resolution of 640 × 640 pixels. The total number of training iterations was 300, with each batch (batch size) processing 16 images. The initial learning rate was set to 0.01, and the seed was set to 42. Additionally, no pre-trained model weight files were utilized during training.

E. EVALUATION METRICS

To validate the effectiveness of the method, this study employs precision (P), recall (R), mean average precision (mAP), and frames per second (FPS) as evaluation metrics. These metrics are used to comprehensively assess the model's detection accuracy and processing speed.

F. EXPERIMENTAL METHODOLOGY

The online detection system for cracked eggs based on an enhanced YOLOv8 and CCD camera comprises three primary steps. Firstly, images of intact eggs and cracked eggs are captured and pre-processed, with the dataset splitting according to the model. Subsequently, a neural network model is selected for training. Considering the requirements for detection accuracy and real-time performance, the currently most popular YOLOv8 object detection algorithm is chosen as the foundation and subsequently refined. Improvements were made to this baseline model. The model's performance was evaluated through comparing Precision-Recall (P-R) curves on the test set and conducting ablation experiments, as illustrated in Figure 6. Finally, both the original YOLOv8 model and our improved cracked egg detection model were deployed onto the device. This enabled online, real-time detection of cracked eggs using the CCD camera, allowing for a direct performance comparison between the models, as shown in Figure 7.

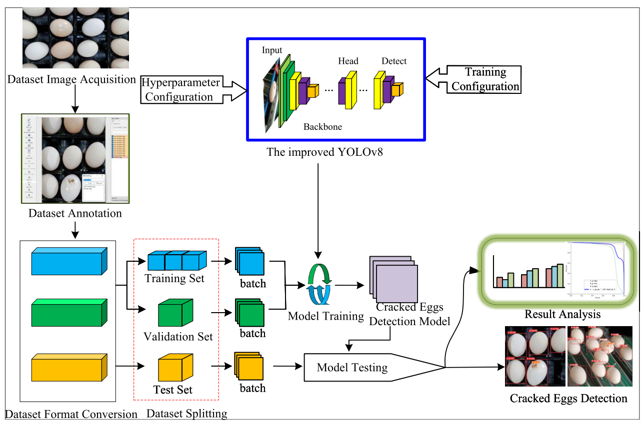


FIGURE 6. Training schematic of the cracked eggs detection model.

III. RESULTS AND ANALYSIS

A. ABLATION EXPERIMENT

An ablation experiment was conducted to validate the effectiveness of the proposed improvements. In this study, the original YOLOv8 was used as the baseline network. The CIoU loss function was replaced with the WIoU loss function, and DCNv2 was incorporated into the backbone network. Furthermore, the BiFPN module was adopted in the neck, and

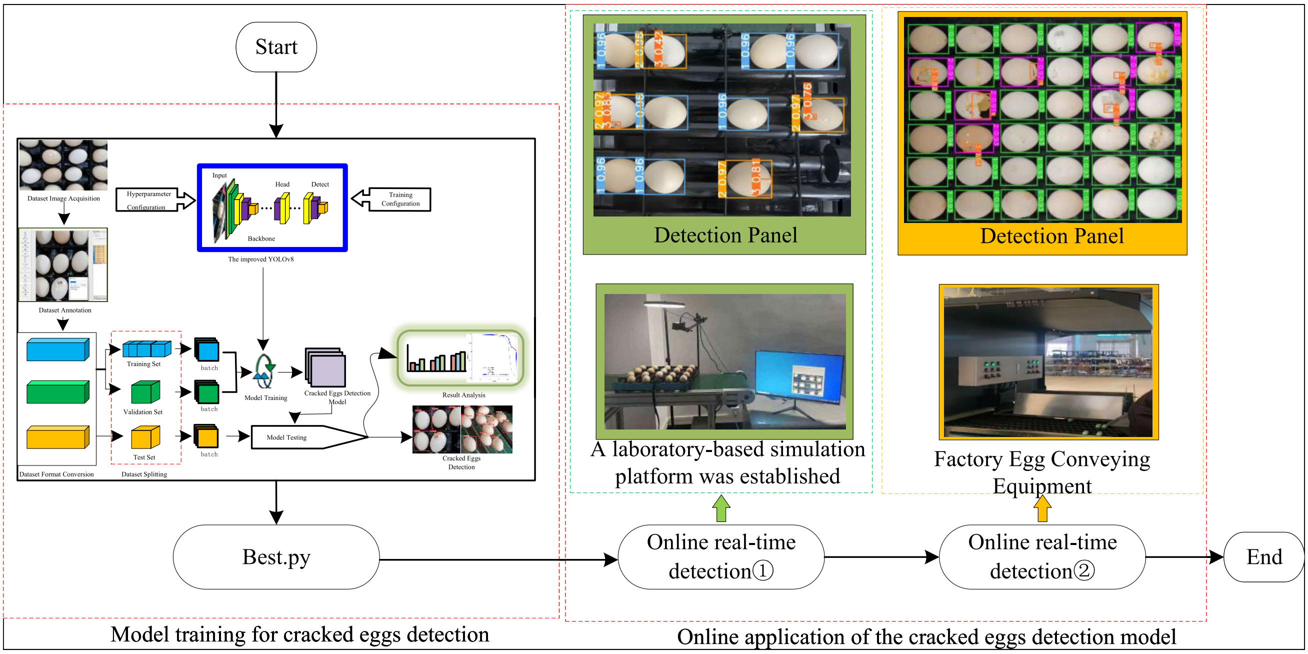


FIGURE 7. Online deployment testing of the cracked eggs detection model.

a small-object detection layer was added to the network. The results of the ablation study are presented in Table 1.

As can be seen from Table 1, all the proposed improvements enhanced the model's performance to varying degrees. The results indicate that replacing the CIoU loss function with WIoU led to a 2.6% increase in the model's mean Average Precision (mAP). The incorporation of the small-object detection layer and the BiFPN module resulted in more substantial gains, improving mAP by 5.3% and 4.4%, respectively. When all four improvement strategies were combined, the model achieved an overall mAP increase of 5.2%. Although the variant with only the small-target head added achieved the highest mAP of 93.0, the model selection criterion is not based solely on a single mAP metric, but rather on a comprehensive trade-off between overall detection accuracy, precision, and robustness. The ensemble model (DCNv2 + BiFPN + WIoU + small-target head) integrates these four enhancements, each of which improves performance at different levels. In particular, this model not only demonstrates excellent performance in terms of mAP (mAP = 92.9) but also achieves the highest precision (P = 94.1%). Therefore, the study selected the ensemble model (DCNv2 + BiFPN + WIoU + small-target head) as the final model to more accurately extract crack-related features from egg images.

Given that the model will ultimately be deployed on production equipment, this study conducted comparative experiments on the original YOLOv8 algorithm and the improved algorithm proposed in this paper, evaluating them across various complexity metrics such as the number of parameters, floating-point operations (FLOPs), and model size (MB). The experimental results are detailed in Table 2.

TABLE 1. Ablation experiment.

DCNv2	BiFPN	WIoU	Small target	Precision (%)	Recall (%)	mAP:0.5 (%)
-	-	-	-	92.2	85.3	87.7
-	-	√	-	90.6	87.4	90.3
√	-	-	-	89.8	83.8	88.2
-	-	-	√	93.8	90.3	93.0
-	√	-	-	93.1	89.4	92.1
-	√	-	√	92.6	89.5	92.2
√	√	-	√	93.8	89.9	92.6
√	√	√	√	94.1	90.0	92.9

TABLE 2. Comparison of experimental results of different network models.

Model	Parameters/10^7	FLOPs/G	Model size/MB
YOLOv8	2.59	79.1	49.6
Our Model	2.66	86.5	51.1

B. COMPARATIVE EXPERIMENTS WITH DIFFERENT DETECTION MODELS

To further evaluate the performance of the improved algorithm in detecting cracked eggs, this study trained Faster RCNN, YOLOv5, YOLOv8, and YOLOv11 on the same dataset and conducted a comparative analysis. The models were compared using precision (P), recall (R), mean average precision (mAP), and frames per second (FPS) as performance metrics. In this experiment, the Faster RCNN model utilized ResNet50 as the backbone, was trained for 300 epochs, and employed pretrained weights. The comparative results are presented in Table 3. It can be observed that compared to the original model, the improved YOLOv8 achieved the highest accuracy on the same dataset, with a mAP of 92.9%. Due to the addition of the small-object detection layer and the BiFPN module, the size of the improved model increased, and its detection speed decreased by 14.5%, resulting in an FPS of 45. Nevertheless, this frame rate still meets the requirements for real-time detection, making the model more suitable for deployment in practical production lines for cracked egg detection. It should be noted that in this experiment, Faster R-CNN employed fine-tuning of pre-trained weights, while the improved YOLOv8 was trained from scratch. This configuration gave Faster R-CNN a head start in the feature extraction phase. Even so, the proposed model still achieved superior performance, further highlighting the robustness of the proposed improvements.

TABLE 3. Comparison of experimental results of different network models.

Model	Precision (%)	Recall (%)	mAP:0.5(%)	FPS
Faster RCNN	68.4	80.7	78.1	17
YOLOv5	86.9	81.1	85.2	58
YOLOv8	92.2	85.3	87.7	52
YOLOv11	92.7	84.3	87.3	47
Our Model	94.1	90.0	92.9	45

C. COMPARISON OF PRACTICAL DETECTION PERFORMANCE

To thoroughly validate the practicality of the improved YOLOv8 algorithm, the trained cracked egg detection model was deployed onto the device for online detection experiments. Initially, a comparative analysis of the detection accuracy between the original and improved models was conducted on a simulated experimental platform set up in the laboratory. Subsequently, the model was tested on a factory conveyor device to evaluate its performance under real-world conditions.

Online detection of cracked eggs was performed using both the original YOLOv8 and the improved YOLOv8 on the experimental platform to assess the performance of the improved YOLOv8 model in practical dynamic detection scenarios. During the dynamic detection process, it was readily observed that the improved YOLOv8 model achieved higher detection accuracy. To facilitate a clear demonstration of the detection effects before and after the model improvements in this paper, a CCD camera was used to record a video of eggs moving on the experimental platform. Subsequently, frames were extracted from this video, resulting in 115 images. These images were annotated according to the method described in Section II-B2 to create a new test set. Finally, the best.pt weight files generated by training the original and improved models were used to run the test.py script, yielding the detection results on the new test set, as shown in Figure 8 and Figure 9.

Figure 8 displays the Precision-Recall (PR) curves obtained from evaluating the original and improved models on the new test set. It can be observed that the improved YOLOv8 achieves a 2.5% higher mean Average Precision (mAP) value compared to the original YOLOv8 on this set. In Figure 9, red bounding boxes represent intact eggs, pink boxes indicate cracked eggs, and brown boxes represent the damaged areas. The results demonstrate that both the original and the improved YOLOv8 models accurately detect intact eggs. For cracked eggs with large damaged areas, both models yield satisfactory detection results. However, when the damaged regions are small, the improved YOLOv8 exhibits significantly higher detection accuracy for cracked eggs than the original model. Furthermore, the improved model also demonstrates better performance in detecting the damaged area. The above experimental results show that the improved YOLOv8 model possesses stronger feature extraction and fusion capabilities, leading to higher overall accuracy in egg identification and superior performance in detecting cracked eggs.

D. PRACTICAL TESTING ON FACTORY EQUIPMENT

To ensure the successful application of the detection model on actual production lines, the study deployed the investigated cracked egg detection model on the egg conveyor line (Kyowa/FP-600) at a local cooperative egg-producing enterprise for online detection, with the conveyor belt operating at 0.1 m/s, a camera model A3200CU000 (1920×1080) capturing images at a frame rate of 120 fps. The system device's computer hardware configuration is shown in Table 4. Similar to the experimental platform set up in the laboratory, the eggs on the Kyowa/FP-600 equipment also do a composite motion of "translation + rotation." This rotation, induced by the conveyor's rollers, ensures that even if the damaged area is initially on the bottom of an egg, it will be rotated into the camera's field of view, thereby enhancing detection accuracy.

TABLE 4. System device configuration.

Category	Specific parameters
Hardware Configuration	CPU: 12th Gen Intel i5-12400F (2.50GHz) GPU: GeForce RTX 3060 Ti 8G Memory: 16GB
Operating System	Windows 10
Software Testing Environment	Python 3.8, PyTorch 1.13.1, CUDA 11.6, cuDNN 8.302

In practical engineering applications, the False Negative Rate (FNR) and False Positive Rate (FPR) are the most critical metrics. The FNR (or missed detection rate) reflects the proportion of target eggs that the model fails to identify, while the FPR (or false detection rate) indicates the proportion of non-target eggs that are incorrectly classified as targets. These two metrics directly determine the reliability of the cracked egg detection system and its practical value in real-world scenarios. Since the original and improved models, under identical software and hardware conditions, detection frames at different rates when inspecting eggs moving at the same conveyor speed, the resulting frame count for online detection is unequal. Therefore, to enable a more accurate and direct comparison of online detection

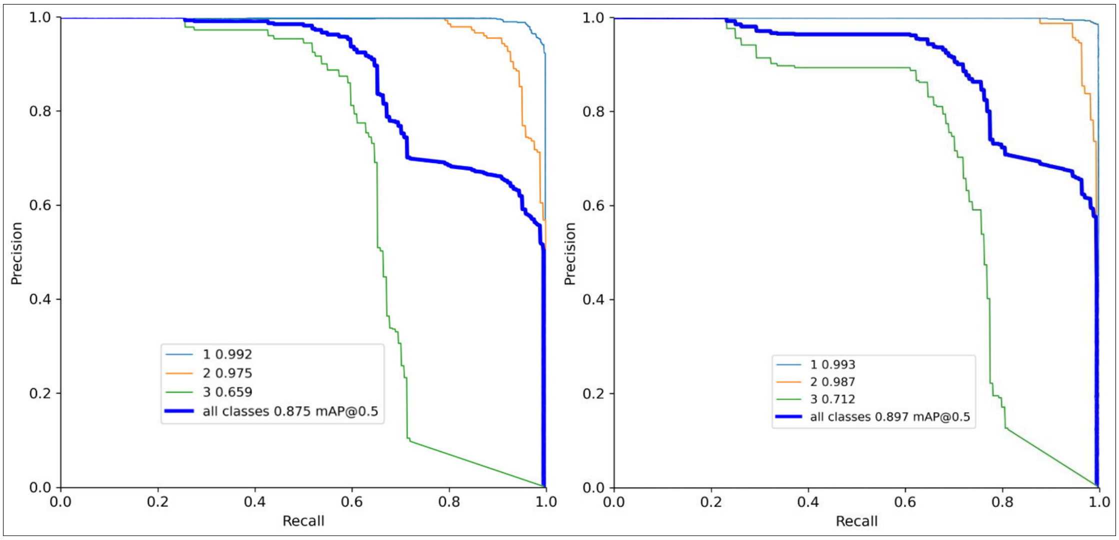
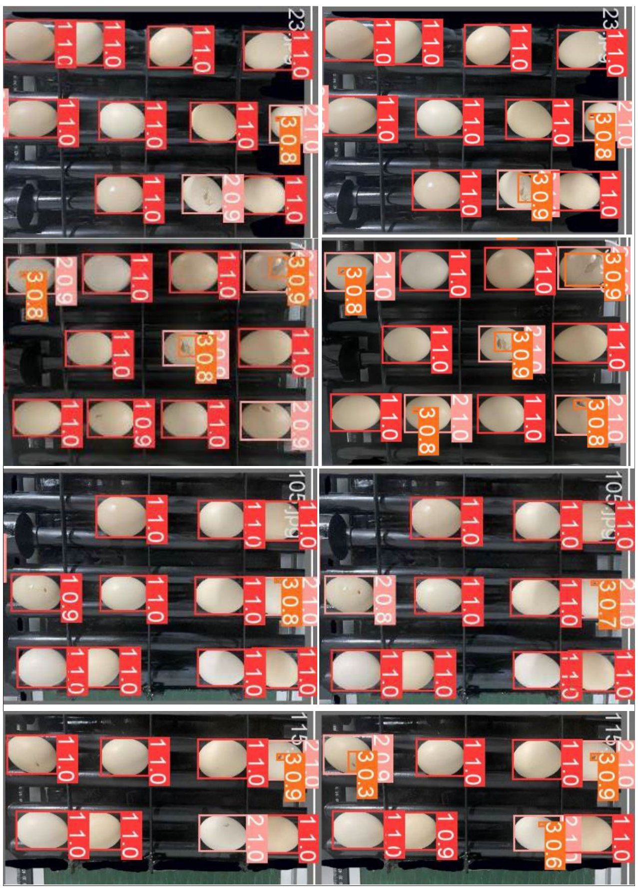


FIGURE 8. **The accuracy of online detection before and after model improvement.**



(a) YOLOv8 model detection results (b) Improved YOLOv8 detection results

FIGURE 9. **Comparison of online detection effects before and after model improvement.**

accuracy before and after model improvement, this study adopted the following procedure: a video of eggs moving on the conveyor was captured using a CCD camera. Frames were then extracted from this video, resulting in 106 images. These images were labelled according to the method described in Section II-B2 to create a new test set. Finally, the best.pt weight files generated from training both the original and improved models were used to execute the detect.py script, yielding the detection results on this new test set. The detection results were then compared with the ground truth annotations to count the number of false negatives and false positives for the three categories: intact eggs, cracked eggs, and damaged areas. And the false negative rate and false positive rate for three categories-intact eggs, cracked eggs, and damaged areas-were calculated, as shown in Tables 5 and 6. The detection results are shown in Figure 10.

TABLE 5. **The accuracy of online detection before model improvement.**

Class	Actual number	V8DN[1]	FNN[2]	FPN[3]	FNR[4] (%)	FPR[5] (%)
1	1746	1794	22	70	1.26	4.00
2	688	640	70	22	10.17	3.20
3	688	536	158	6	22.97	0.87

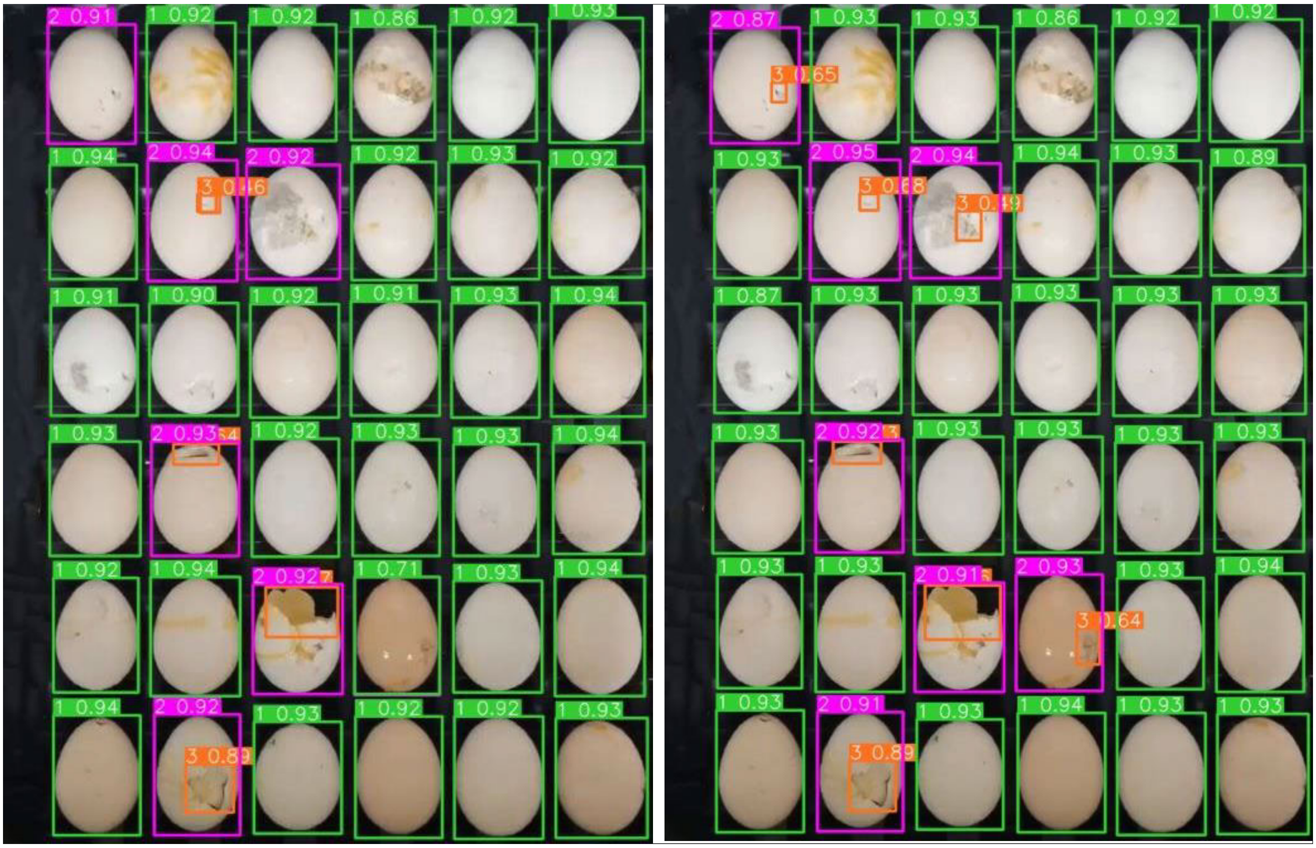
[a] Number of YOLOv8 detections. [2] Number of false negative. [3] Number of false positive. [4] False negative rate. [5] False positive rates.

TABLE 6. **The accuracy of online detection after model improvement.**

Class	Actual number	V8DN[1]	FNN[2]	FPN[3]	FNR[4] (%)	FPR[5] (%)
1	1746	1776	8	38	0.46	2.18
2	688	658	38	8	5.52	1.16
3	688	560	132	4	19.19	0.58

[a] Number of improved YOLOv8 detections. [2] Number of false negative. [3] Number of false positive. [4] False negative rate. [5] False positive rates.

As illustrated in Tables 5 and 6, the improved YOLOv8model demonstrates a substantial reduction in both false negative and false positive rates across all three



(a) YOLOv8 model detection results

(b) Improved YOLOv8 detection results

FIGURE 10. Detection effect on the actual production line.

categories—intact eggs, cracked eggs, and damaged areas—compared to the original model. Specifically, the false negative rates decreased by 0.80%, 4.65%, and 3.78%. Respectively, while the false positive rates were reduced by 1.82%, 2.04%, and 0.29%, respectively. These results indicate a comprehensive enhancement in the model's detection capability, with the most notable improvement observed in the recognition of cracked eggs. In Figure 10, green bounding boxes represent intact eggs, purple boxes denote cracked eggs, and brown boxes represent the damaged areas. It can be observed from Figure 10 that both the original and improved YOLOv8 models can accurately detect intact eggs. However, for cracked eggs and damaged regions, the original YOLOv8 exhibits a higher false negative rate, whereas the improved model shows significantly superior performance in these challenging cases. The final experimental results demonstrate that the proposed improved model significantly enhances the identification and localization accuracy for the regions of interest within the target detection task. Under the dual verification mechanism, both Category 2 and Category 3 detect whether an egg is broken. A cracked egg is only considered missed when both Category 2 and Category 3 fail to detect it simultaneously. As shown in Table 6, the data on false positive and false negative rates for broken eggs and damaged areas indicates that the improved YOLOv8 reduces the false negative rate for broken eggs (where neither the egg breakage nor the damaged area is detected) to 1.06%($5.52\% \times 19.19\% = 1.06\%$), while keeping the false positive rate for broken eggs below 2%(false acceptance rates of 1.16% and 0.58% for cracked eggs categories 2 and 3).

Figure 11 illustrates a comparison of the online detection speed between the original and improved YOLOv8 models. During online detection, the inference time for each frame was recorded. The inference times for the first 400 frames were then converted into the frame rate (FPS) and stored in a.csv file, which was used to plot the FPS comparison graph. The results demonstrate that the detection frame rates of both the original and improved models meet the requirements for real-time online detection, which was also confirmed by the visual output on the display. Specifically, the original model achieved an average online detection frame rate of approximately 52 FPS, while the improved model achieved an average rate of about 45 FPS, which is slightly lower than that of the original.

IV. DISCUSSION

This study successfully developed and validated an improved YOLOv8-based model for cracked egg detection tailored for industrial production lines. The experimental results

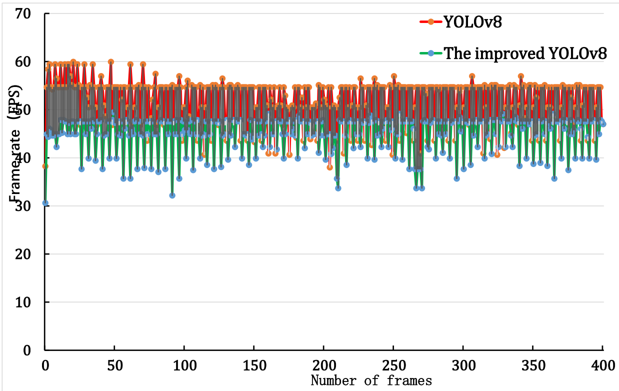


FIGURE 11. Speed of online detection before and after model improvement.

demonstrate that the series of enhancement strategies we introduced, including BiFPN for multi-scale feature fusion, DCNv2 for geometric adaptation to irregular cracks, WIoUv2 for optimizing hard examples, and the addition of a small-object detection layer-effectively worked in synergy, significantly improving the overall performance of the model. The final performance of a model requires a balance between accuracy and speed. As shown in Table 3, our improved model significantly outperformed mainstream models such as Faster R-CNN, YOLOv5, and YOLOv11 in terms of mAP. Although the detection speed (45 FPS) is slightly lower than that of the original YOLOv8 (52 FPS) due to increased model complexity, it still far exceeds the requirement for real-time detection on production lines ($\geq$ 30 FPS). This trade-off, accepting a modest speed reduction in exchange for a substantial gain in detection accuracy, is crucial and highly valuable for industrial applications focused on ensuring product quality and reducing waste.

Compared to recent studies focusing on improving YOLO models for egg inspection [30], [31], our research offers an end-to-end solution tailored for real-world dynamic scenarios. While prior work primarily optimized model accuracy on static datasets, our approach demonstrates robustness against motion blur, lighting variations, and other dynamic disturbances through deployment on both simulation platforms and actual production lines (Kyowa/FP-600). More importantly, our proposed dual-verification mechanism for cracked eggs and damaged areas, combined with the production line's inherent egg rotation device, establishes a reliable detection and sorting decision process. This process reduces the false-negative rate for cracked eggs by 4.65% (Tables 5 and 6).

It should also be noted that eggs tested on industrial production lines in this study were unwashed, whereas eggs used in laboratory platform tests were purchased directly from supermarkets and were pre-washed. Experimental results demonstrate that the proposed model is effective not only for unwashed eggs but also for detecting cracks in pre-washed eggs. Although the model demonstrates good adaptability to both types of eggs, its generalization capabilities for more diverse damage types (e.g., reticular cracks) and different poultry egg varieties remain under-validated. For eggs with a uniform, solid-colored shell, such as duck eggs, it is anticipated that the method will maintain relatively strong performance. This is because the visual cues of damage, primarily abrupt changes in color and surface glossiness, are prominent features that the model can effectively learn during training. Conversely, for eggs with intricate patterns, like quail eggs, the situation is twofold: the intact shell presents a complex background, but once breakage occurs and yolk leakage happens, the distinctive 'wet and reflective' characteristic of the damaged area still provides a reliable signal for detection. A more significant challenge lies in identifying complex crack patterns, such as fine 'net-like' cracks. These defects typically occupy a small pixel area in the image, exhibit weak edge gradient information, and are easily confused with natural shell textures or minor stains, particularly on unwashed eggs. This ambiguity can substantially increase the difficulty of recognition, potentially leading to a higher false negative rate. Addressing this specific challenge by collecting more diverse samples of complex cracks and designing more nuanced feature extraction mechanisms will be a key focus of future work. It is suggested that subsequent research may adopt novel methods such as dark-box illumination to enhance the visibility of crack features.

Future research will focus on several directions: First, exploring more advanced small object detection algorithms, such as feature super-resolution or attention-guided magnification mechanisms, to specifically reduce the false negative rate in damaged areas. Second, we plan to collect datasets featuring more diverse damage patterns and different poultry egg varieties to further test and enhance the model's generalization capabilities. Finally, we consider integrating the system into a broader poultry egg quality inspection framework to achieve unified detection of internal defects such as scattered yolk and blood spots.

V. CONCLUSION

This study aims to address the challenge of real-time, precise detection of cracked eggs on egg production lines. To this end, we propose an improved detection model based on YOLOv8 and establish a comprehensive technical workflow spanning data collection, model refinement, and production line deployment.

(a) This study innovatively introduces a dual-verification, refined detection method. As long as the model detects either a "cracked egg" or a "crack region", the system classifies the egg as cracked. By separately learning features of localized damage areas, the model enhances its perception and generalization capabilities for various damage patterns, improving detection robustness against unseen samples and complex environments.

(b) Through structural improvements such as the introduction of BiFPN, DCNv2, WIoUv2, and a small target detection layer, the model's ability to perceive cracked areas, multiscale feature fusion, and handling difficult cases is significantly enhanced. Offline experiments demonstrate that on the test dataset, the model achieves a mAP of 92.9%, representing a 5.2% improvement over the original YOLOv8 and outperforming other comparison models.

(c) This study deployed an improved version of the YOLOv8 model on an egg conveyor line and implemented end-to-end online detection using a Python inference framework. Test results showed a high accuracy rate. The false negative rate for intact eggs was 0.46%, and the false positive rate was 2.18%, representing an improvement over the 4.00% false positive rate observed before model optimization. After implementing a dual verification mechanism, the false negative rate for cracked eggs was approximately 1%, and the false positive rate was approximately 2%. Furthermore, the model achieved real-time detection at a frame rate of 45 frames per second, with accuracy and operational speed meeting industrial requirements.

In summary, this study provides a reliable and efficient solution for automated quality inspection on egg production lines, playing a positive role in enhancing the intelligence level and economic benefits of the poultry egg industry.

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